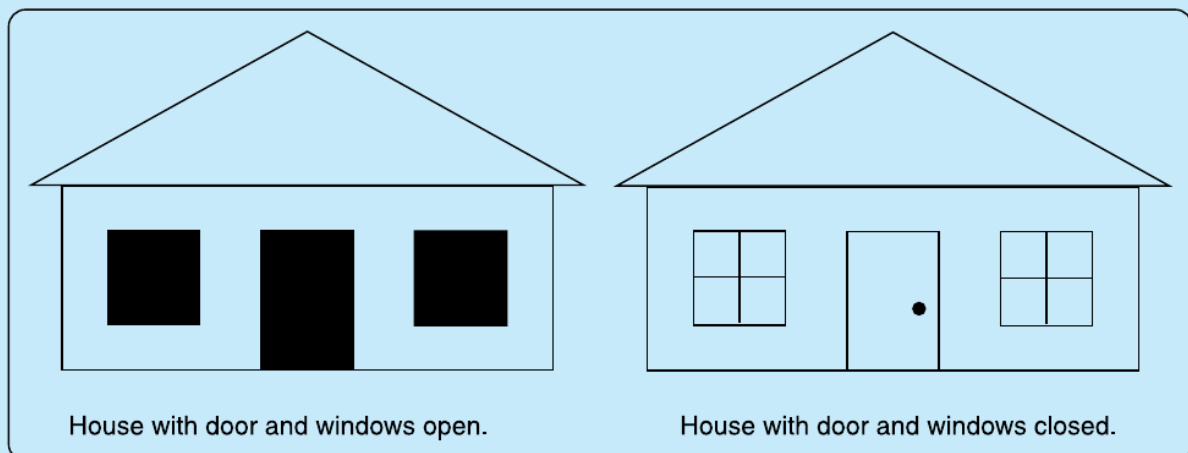


Programming Challenges

1. This Old House

Using basic JavaFX shape nodes (Rectangle, Polygon, Circle, and so on), draw a house with at least two windows and a door. Feel free to add extra touches such as a sky, sun, or clouds. As an added interaction, make each door and window clickable: clicking one should toggle it between “open” (outlined) and “closed” (filled in).

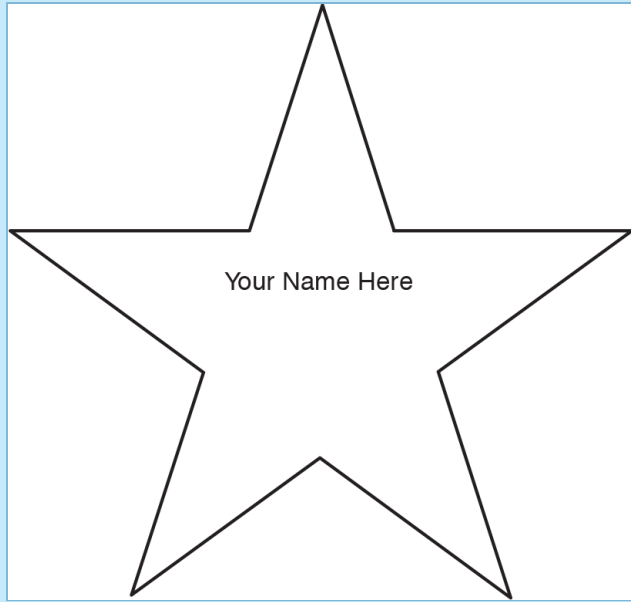


2. Tree Age

Counting a tree's growth rings is a good way to estimate its age; each ring represents one year. Using Circle or Ellipse shapes, draw the growth rings of a five-year-old tree, then use the Text shape to number each ring, starting at the center and working outward.

3. Hollywood Star

Design your own star for the Hollywood Walk of Fame. Write a JavaFX application that draws a five-pointed star (using a Polygon) with your name displayed inside it using a Text node.



4. Vehicle Outline

Using the shape nodes covered in this chapter, draw the outline of a vehicle of your choice (car, truck, airplane, etc.).

5. Solar System

Use the `Circle` class to draw each planet of the solar system. Draw the sun first, then each planet in order of distance from the sun (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, and the dwarf planet Pluto). Label each body with a `Text` node.

6. Opacity Adjuster

Build a JavaFX application that lets the user select an image file via a `FileChooser` dialog, then displays it in an `ImageView` alongside a `Slider` that adjusts the image's opacity in real time.

7. Color Adjuster

Build a JavaFX application that lets the user select an image file via a `FileChooser` dialog, displays it in an `ImageView`, and provides four `Slider` controls (using a `ColorAdjust` effect) that let the user adjust the image's hue, saturation, brightness, and contrast.

8. Mouse Rollover

Build a JavaFX application that displays an image. When the mouse pointer moves over it, swap in a second image; the second image should remain displayed as long as the pointer stays over it, then revert to the first image once the pointer moves away.

9. Coin Toss

Build a JavaFX application that simulates a coin toss. The program should wait for a key press, then generate a random number in the range 0–1. Display the image of a coin showing heads if the number is 0, or tails if it is 1.

10. Lunar Lander

Using a spacecraft image and a background image of the lunar surface, build a JavaFX application that initially shows the spacecraft resting on the surface. When the user presses the spacebar, animate the spacecraft (for example with a `TranslateTransition`) so it slowly lifts off.

11. Change for a Dollar Game

Using images of a penny, nickel, dime, and quarter, build a JavaFX game that displays all four coins plus a “Count Change” text element. The user may click any coin, in any order, any number of times. Clicking “Count Change” should total the value of everything clicked and announce whether it comes to exactly one dollar — if so, the player wins.

12. Rock, Paper, Scissors Game

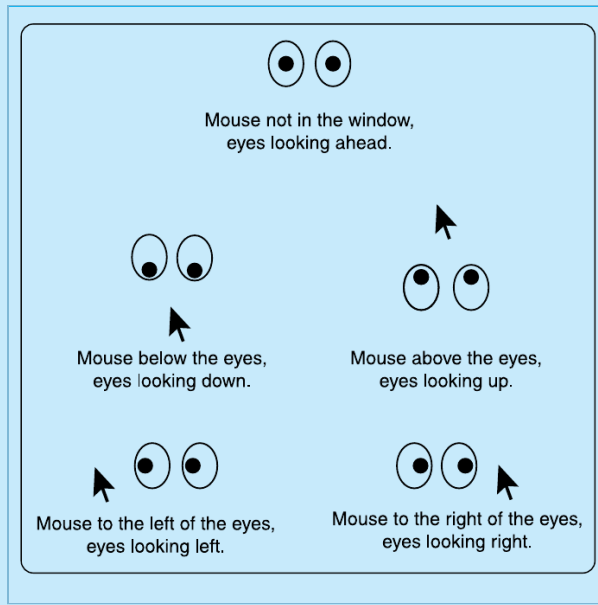
Build a JavaFX application that lets the user play rock-paper-scissors against the computer. On start, generate a random number 1–3 (1 = rock, 2 = paper, 3 = scissors) to represent the computer's hidden choice. The user picks by clicking an image/Button, and their choice appears in an `ImageView`. Reveal the computer's choice, then determine the winner: rock beats scissors, scissors beats paper, paper beats rock; matching choices are a tie and should be replayed.

13. FollowMe

Build a JavaFX application that initially displays the word “Hello” centered in the window. As the user moves the mouse, the text should follow the cursor around the scene.

14. WatchMe (Eyes That Follow)

Build a JavaFX application that draws two eyes in the center of the window, each with a `Circle` pupil. While the mouse pointer is outside the window, the eyes should look straight ahead. While the pointer is inside the window, the pupils should track its position.



15. Thermometer

Build a JavaFX application that draws a thermometer using basic shapes. Provide a Slider that the user can drag to set a temperature; as the slider moves, update the thermometer's fill level to match.

16. Polygon Drawer

Build a JavaFX application that lets the user click six points anywhere in the window. After the sixth click, connect the points into a closed Polygon with a vertex at each clicked location.

17. GridFiller

Build a JavaFX application that displays a 4×4 grid (for example, using GridPane). Clicking an empty cell should draw a filled circle in it; clicking a cell that already contains a circle should remove it.

18. Stopwatch

Build a JavaFX application that simulates a stopwatch, with Start and Stop buttons. Clicking Start should begin counting elapsed seconds in a Label (for example, using a Timeline or AnimationTimer); clicking Stop should freeze the count.

19. Slideshow Application

Build a JavaFX application that runs a slideshow of images with a timed delay between each one. Let the user select up to ten images (via a FileChooser) and specify the delay, in seconds, before advancing to the next image.

20. Card Dealer

Using a set of images for a complete deck of poker cards, build a JavaFX application with a “Deal a Card” Button. Each click should randomly select a card still remaining in the deck and display it; once dealt, a card is removed from the deck and cannot be selected again. When the deck is empty, display a message saying so.

